

Caoliwen Wang

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Research Interests

Rooted in computer vision and graphics, scientific computing, and generative AI, my research develops **physically grounded, scalable, and generative models** for physical world simulation and video/3D content generation.

Education

University of British Columbia

Sep 2025 - Present

Ph.D. in Computer Science

Advisor: Prof. Peter Yichen Chen

University of Science and Technology of China

Sep 2021 - Jun 2025

B.S. in Computational Mathematics

Advisor: Prof. Ligang Liu

Research Experience

Scalable Physical Simulation

Apr 2025 - Present

- **WorldParticle:** Developed a unified model that represents diverse physical phenomena as Lagrangian particles and learns their dynamics with a Transformer-based backbone, enabling fully data-driven simulation without the hand-crafted design and pre-defined physical biases required by traditional solvers, toward unified and scalable world modeling.
- **WorldParticle++:** Built upon the WorldParticle backbone, scaling particle-based simulation to million-scale systems with support for dynamic boundaries, while leveraging forcing techniques from video-based diffusion models to improve rollout stability for long-horizon simulation.
- **Agentic Simulation:** Developed a fully automated simulation pipeline leveraging LLM agents for 3D scene construction and solver configuration, eliminating tedious manual parameter tuning, making large-scale 3D data generation feasible, and enabling integration with WorldParticle toward a unified 3D Physics Foundation Model.
- **Agentic Sim-to-Real:** Leveraged LLM agents to automatically close the sim-to-real gap for robotic manipulation of diverse types of powders, generating scalable data for training policies that generalize across different powder materials.

Physics Informed Video Generation

May 2026 - Present

- **Material-aware Video Generation:** Fine-tuned Wan2.2 with multi-material and non-interaction video data, demonstrating that material-aware training can significantly enhance video foundation models' ability to generate videos involving interactions among diverse materials.

Physically Plausible 3D Generation

Apr 2025 - Oct 2025

- **Gauss-Seidel Projection:** Developed a method that enforces physical validity, including non-penetration and structural plausibility, on generated 3D protein configurations, improving the inference efficiency of protein structure prediction foundation models.

Design-Driven 3D Shape Generation

Jan 2024 - Jun 2025

- **Neural Shadow Art:** Developed an implicit neural 3D representation that facilitates geometry generation from multi-view binary shadow targets, with angular adjustment and volume/geometry optimization.
- **Shape from Semantics:** Developed a 3D generation framework for creating semantically consistent shapes whose multi-view renderings align with user-specified prompts, extending the inverse-modeling framework to a semantics-conditioned 3D generation problem.

Industry Experience

Style3D

Jun 2026 - Present

- **WorldParticle Scaling:** Scaled WorldParticle to human motion and cloth grasping scenarios, demonstrating its ability to generalize across diverse human motions and cloth manipulation strategies.

Publications & Preprints

- [1] **Caoliwen Wang**^{*}, Minghao Guo^{*†}, Siyuan Chen^{*}, Heng Zhang, Mengdi Wang, Xingyu Ni, Hanson Sun, Kunyi Wang, Zherong Pan, Kui Wu, Lingjie Liu, Yin Yang, Chenfanfu Jiang, Taku Komura, Wojciech Matusik, and Peter Yichen Chen[†]. **WorldParticle: Unified World Simulation of Lagrangian Particle Dynamics via Transformer**. *SIGGRAPH Asia (Conference Track)*, 2026.
- [2] Xingyu Ni[†], Jiong Chen, Siyuan Chen, **Caoliwen Wang**, Mathieu Desbrun, and Taku Komura. **Efficient Open-Boundary Poisson Solves**. *ACM Transactions on Graphics (Proc. SIGGRAPH Asia)*, 2026.
- [3] Siyuan Chen^{*}, Minghao Guo^{*†}, **Caoliwen Wang**, Anka He Chen, Yikun Zhang, Jingjing Chai, Yin Yang, Wojciech Matusik, and Peter Yichen Chen[†]. **Physically Valid Biomolecular Interaction Modeling with Gauss-Seidel Projection**. *International Conference on Learning Representations (ICLR)*, 2026.
- [4] Liangchen Li, **Caoliwen Wang**, Yuqi Zhou, Bailin Deng, and Juyong Zhang[†]. **Shape from Semantics: 3D Shape Generation from Multi-View Semantics**. *Preprint, Under Review at TVCG*, 2025.
- [5] **Caoliwen Wang**, Bailin Deng, and Juyong Zhang[†]. **Neural Shadow Art**. *Pacific Graphics (Conference Track)*, 2025.

Awards

University and Provincial Outstanding Graduate Award 2025 (USTC)

Reviewing

ACM SIGGRAPH Asia, Pacific Graphics